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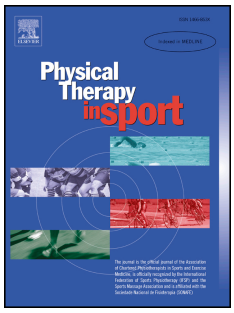

The British Athletics Muscle Injury Classification grading system as a predictor of return to play following hamstrings injury in professional football players

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Title: The British Athletics Muscle Injury Classification grading system as a predictor of return to play training following hamstrings injury in professional football players.

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Keywords: Hamstrings, Hamstrings Strain Injury, MRI, return to play, football.

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ABSTRACT

Objectives: Investigate the British Athletics Muscle Injury Classification (BAMIC) grading system as a predictor of return to play (RTP) following primary hamstring strain injury (HSI) and its agreement with the Peetron's classification system in professional footballers.

Methods: A retrospective cohort study of 39 hamstrings strains in a professional English football club were identified. Two musculoskeletal radiologists reviewed historical MRI's and classified them against the BAMIC and Peetron's grading system. Classification, oedema length and cross-sectional area were compared against RTP.

Results: Pearson's correlation coefficient demonstrated a weak but statistically significant correlation between BAMIC and RTP (r=0.32; 95%CI 0.01 to 0.58; p=0.05). Maximum length of intramuscular oedema demonstrated weak correlations with RTP (r=0.3; 95%CI -0.02 to 0.56; p=0.06). Percentage cross sectional demonstrated a weak correlation with RTP (r=0.02; 95%CI -0.3 to 0.33; p=0.91). Multiple regression demonstrated that 16% of the variance in RTP was explained by the model. Kappa for the agreement between BAMIC and Peetron's was 0.21 (95%CI 0 to 0.42).

Conclusions: A significant association between the grade of HSI on the BAMIC system and RTP was found. Findings suggest BAMIC could provide valuable prognostic information on the RTP.

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Introduction

Hamstring strain injuries (HSI) remain the most common muscle related injury in professional football (Ekstrand *et al.* 2021). Despite efforts to reduce the number of HSI through a range of reduction strategies (Chesterton *et al.* 2021; Chesterton *et al.* 2022), Ekstrand, Walden and Hagglund (2016) reported an annual increase of 2.3% in HSI incidence in professional football over a 12-year period. Furthermore, time-lost from competition for a HSI is on average 18-19 days (Woods *et al.* 2004, Ekstrand, Hagglund & Walden, 2016; Ekstrand *et al.* 2020; Shamji *et al.* 2021) with reoccurrence rates ranging from 12-33% (Woods *et al.* 2004; Hagglund, Walden & Ekstrand, 2009; Ekstrand, Hagglund and Walden, 2011; Ekstrand *et al.* 2020) indicating a significant injury burden.

Hamstrings strain injuries provide a significant challenge to medical teams in both the management and accurate estimation of a safe and efficient return to play (RTP) timeframe (Moen *et al.* 2014; Hamilton *et al.* 2018). To facilitate this process many medical teams, employ magnetic resonance imaging (MRI) to provide a HSI diagnosis and grade (Reurink *et al.* 2015). Despite being accepted as the most accurate and sensitive method for assessing muscle injuries (Connell *et al.* 2004; Rettig, Meyer and Bhadra, 2009; Kerkoffs *et al.* 2013; Crema *et al.* 2016) there is limited evidence to support MRI's role in predicting an athletes' return to participation (Reurink *et al.* 2015). Evidence suggests that complete resolution of a hamstring intramuscular tendon injury on MRI is not considered necessary for a clinically successful RTP (Vermeulen *et al.* 2021).

Previously, muscle injuries were traditionally graded on MRI, using the modified Peetron's classification (Peetron *et al.* 2002). Despite having excellent reliability (Ekstrand *et al.* 2012; Hamilton *et al.* 2014), this ordinal scale has been criticised for providing little representation of the length of the muscle tear, distance from the origin, cross-area of the oedema and involvement of the tendon (Pollock *et al.* 2014; Grassi *et al.* 2016; Hamilton *et al.* 2018; Wangensteen *et al.* 2018). Consequently, alternative grading systems have been established that provide a more detailed classification of muscle injuries. The Munich Muscle

Classification for example, has been widely implemented in professional football for MRI grading of HSI (Mueller-Wohlfahrt *et al.* 2013; Ekstrand *et al.* 2013), but it has been rebuked for having ill-defined structural and functional elements within the classification system (Pollock *et al.* 2014; Hamilton *et al.* 2017). A proposed method by Chan et al (2012), has tried to integrate the location of the injury, defined as the involvement of the proximal or distal musculotendinous junction or muscle body, with a precise three-grade MRI assessment. Moreover, a further sub-classification of injuries directly affecting the muscle body was suggested, specifying proximal, middle or distal location and fascial involvement. However, there is no descriptor for primary tendon injury, no grading for 'MRI negative' injuries and it has been criticised for some of the confusing terminology (Hamilton *et al.* 2017). In English football there has been a shift towards the British Athletics Muscle Injury Classification (BAMIC) system due to its clearly defined anatomically focused grading (MacDonald *et al.* 2019). The BAMIC provides information on the length of oedema, cross sectional area and the location of the injury within the muscle (Pollock *et al.* 2014) and has demonstrated at least 'substantial-to-almost perfect' agreements for intra and inter-rater reliability (Patel *et al.* 2015; Wangensteen *et al.* 2017). However, there is limited data on the implementation of BAMIC in professional football to assist practitioners in this field.

The primary aim of this study was to establish if the BAMIC grading system is a useful predictor of return to play (RTP) following HSI in professional football. Additionally, we aimed to investigate if any of the parameters of the BAMIC, such as cross-sectional area and the length of intramuscular oedema correlates with RTP. Finally, we aimed to investigate the agreement between BAMIC and Peetrons' classification systems, to establish if pre-existing data in this area can inform practitioners utilising the BAMIC in professional sport.

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METHODS

A retrospective observational study was conducted by the medical department at a professional English Football Club over a 9-year period from 2010-2019. All adult male players signed on a professional contract (aged 18 years and over) were eligible for inclusion within the study. XX School of Social Science, Humanities and Law Ethics Committee approved the study, which conformed to the Declaration of Helsinki (SSSBLEC147).

Only hamstrings strain injuries requiring an MRI within 7 days of the initial onset of the clinical symptoms were included in the study. The team physicians and physiotherapist registered all injuries in accordance with the FIFA Medical Research and Assessment (Fuller et al. 2016). Any injuries that did not include a definitive return to training date were excluded along with any injury that could have been classified as a reoccurrence. Reoccurrence of injury was defined in agreement with the classifications reported by Fuller et al (2006) that defined an 'early occurrence' as a muscle injury occurring at the same site within 2 months of the return to sport.

Return to play (RTP), was calculated as the point in time when the player ceased physical activity due to the HSI to when they returned to full participation in team training and were available for match selection (Fuller et al, 2006, Hagglund *et al.* 2005). The medical practitioners were not blinded from the results of the original MRI and were informed of the severity of the hamstrings strain injury. Initial MRI classifications for the purpose of diagnosing injury severity were based upon the modified Peetron's classification (2009-2015) and later BAMIC classification (2015-2019). No specific pre-determined hamstrings return to play protocol was used during this period and no further imaging took place. The decision for RTP was made by the lead physicians involved in the management of the injury. The decision for RTP was multifactorial but was determined by a combination of the player's clinical presentation, rehabilitation progress and subjective readiness to return. Common clinical markers such as cessation of pain, symmetrical flexibility and the return of pre-injury

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strength were used in conjunction with performance markers. Performance data collected via a global positioning system (GPS) monitored variables such as sprint velocity, acceleration and decelerations and the capacity to complete a volume of high-speed running comparable to match-day demands. All these factors were employed to assist in the RTP decision.

Magnetic Resonance Imaging Protocol

39 potential HSI were identified and manually extracted from the electronic medical database. Once the injuries were identified a team of two consultant musculoskeletal radiologists of 12 and 20 years of experience independently reviewed the original MRI images. The radiologists were blinded from all identifying information except clinical data in relation to the HSI.

Standardised measuring protocols were utilised by both consultant radiologists. All images were obtained using a 1.5T MRI scanner. Fluid sensitive fat suppression sequences along with T1 sequences were used. The longitudinal length of each lesion (cm) was recorded. The cross-sectional area was calculated by identifying the MRI sequence with the largest percentage of total muscle involvement in the transverse plane (De Vos et al. 2014). The involved muscle and grade of injury using both the Peetron's and BAMIC systems was recorded. In cases where more than one hamstring was involved the muscle with the most extensive oedema was scored (Hallen & Ekstrand 2014, Shamji *et al.* 2021). Hamilton and colleagues (2014) have previously reported excellent interobserver and intraobserver reliability for MRI grading and prognostic parameters in acute hamstring pathology.

Magnetic Resonance Imaging Grading

Muscles injuries were classified using the BAMIC system and a modified Peetron's classification. BAMIC classification was based on the percentage of cross-sectional area of the muscle involved, the length of the oedema (0-4) and the location in the muscle (a-c) (Pollock *et al.* 2014). In addition, the modified Peetron's classification was used based on the

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following criteria: grade 0 negative MRI without any visible pathology, grade 1 oedema but no architectural distortion, grade 2 architectural disruption indicating partial tear and grade 3 total muscle or tendon rupture (Peetrons, 2002). The addition of the modified Peetron's classification was utilised as a reference point to allow direct comparison with previous RTP studies analysing HSI and to allow analysis between the MRI classification systems to be identified.

Data Analysis

Data was analysed using Microsoft Excel (Microsoft 2016, Microsoft Corp, Redmond, WA) IBM SPSS 26 (IBM 2017**,** IBM Corp, Armonk, NY) statistical software. Descriptive statistics were used to report mean, median, range and standard deviation to establish timeframes with the corresponding injury grading using the modified Peetron's classification and the BAMIC grading systems.

To investigate the effectiveness of BAMIC grading system as a predictor of RTP, the BAMIC score was converted to a 0-9 scale. A linear regression with BAMIC score as the predictor and RTP as the outcome variable was then conducted.

The study also aimed to investigate the agreement between the BAMIC and the Peetron's grading systems. To compare the two methods, only the numerical score was assessed against Peetron's without the addition of the alphabetic score. Agreement was estimated by calculating kappa and was interpreted as; < 0 poor or none, 0 to 0.2 minimal, 0.2 to 0.4 weak, 0.4 to 0.6 poor, 0.6 to 0.8 strong and 0.8 to 1 almost perfect (Munoz and Bangdiwala, 1997; McHugh, 2012;). Finally, we assessed the combination of BAMIC, percentage cross-section of injury and length of oedema as predictors of RTP using a multiple regression (Supplementary Material 1). Assumption tests for multiple regression were conducted and met, as described by Field (2009), including assessment of heteroscedasticity and non-linearity and the normal distribution of residuals.

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RESULTS

39 HSI from 30 professional football players met the studies' criteria (mean age at time of injury = 26.26 ± SD 4.6 years, range 18-35 years). An isolated Biceps Femoris injury was the most commonly injured muscle of the hamstrings (n=26; 66.6%), followed by semimembranosus (n=6; 15.4%) and semitendinosus (n=2; 5%). Injury effecting all the hamstring muscles was reported on 1 occasion (2.6%) and there were 3 incidences of the injury effecting the biceps femoris and semitendinosus (7.7%). One injury (2.6%) was reported as a BAMIC 0a and the muscle group was recorded as 'no injury'. The overall mean for RTP was 16.5 (SD 12.2, median 12, range 2-56) days. The RTP for each of the injury classification are presented in table 1.

Table 1 – Hamstrings Strain Injury Classification, Grade and Mean Time to Return to Play

MRI Grading System	Muscle Injury Classification	No. of Injuries (percentage)	Mean RTP days (SD)
Modified Peetrons	Grade 0	2 (5.1%)	5.5 (0.5)
	Grade 1	18 (46.1%)	15.3 (12.2)
	Grade 2	18 (46.1%)	18.9 (12.3)
	Grade 3	1 (2.6%)	15 (0)
BAMIC	0a	1 (2.6%)	6 (0)
	0b	1 (2.6%)	13 (0)
	1a	4 (10.3%)	15.5 (14.9)
	1b	3 (7.7%)	7 (2.2)
	2a	3 (7.7%)	17.67 (8.3)
	2b	18 (46.2%)	14.9 (10.3)
	2c	4 (10.3%)	23.75 (5.8)
	3a	0	0
	3b	3 (7.7%)	25.33 (21)
	3c	2 (5.1%)	23.5 (11.5)

British Athletics Muscle Injury Classification and Return to Play

Pearson's correlation coefficient demonstrated a weak but statistically significant correlation between BAMIC and RTP (r=0.32; 95%CI 0.01 to 0.58; p=0.05). The ANOVA for the overall regression model was statistically significant, F=4.07; df=1; p=0.05. R^2 = 0.1 indicating that 10% of the variance in RTP can be explained by the BAMIC score (Standard error of the

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estimate = 11.88). The Beta statistic for the intercept was 7.23 (95%CI -2.82 to 17.27; standard error = 4.96) and for BAMIC was 1.94 (95%CI -0.01 to 3.88; standard error = 0.96).

Return to Play and Oedema Length/Cross Sectional Area

Maximum length of intramuscular oedema demonstrated weak but not statistically significant correlations with RTP (r=0.3; 95%CI -0.02 to 0.56; p=0.06). Percentage cross section also demonstrated a weak and non-statistically significant correlation with RTP (r=0.02; 95%CI -0.3 to 0.33; p=0.91). The ANOVA for the overall model was non-significant, (F=2.2; df=3; p=0.1; R^2 = 0.16), indicating that 16% of the variation in RTP is explained by the model (standard error of the estimate =11.8). Beta statistics and their 95%CI are presented in table 2.

Table 2 - Beta statistics for multiple regression

Predictor	Beta	Standard Error	95%CI	Standardised Beta
Intercept	8.54	5.28	-2.18 to 19.26	-
BAMIC	1.18	1.11	-1.08 to 3.44	0.19
Percentage cross section	-0.14	0.12	-0.38 to 0.1	-0.23
Length of oedema	0.87	0.56	-0.28 to 2.01	0.34

All predictors were non-significant (p>0.05)

Agreement between British Athletics Muscle Injury Classification and modified Peetron's Classification

There was insufficient data to achieve a reliable indication of agreement between BAMIC and Peetron's, (table 3). There were numerous data points which had expected counts of less than 5. Kappa for the agreement between the two scoring methods was 0.21 (95%CI 0.00 to 0.42) (weak agreement).

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Table 3 - Cross-Tabs Table for Agreement between Peetron's and BAMIC Scores

			BAMIC Score				
			0	1	2	3	Total
Peetron's score	0	Count	1	1	0	0	2
		Exp. Count*	0.1	0.4	1.3	0.3	2
		% Total	2.6	2.6	0	0	5.1
	1	Count	1	6	11	0	18
		Exp. count	0.9	3.2	11.5	2.3	18
		% Total	2.6	15.4	28.2	0	46.2
	2	Count	0	0	13	5	18
		Exp. count	0.9	3.2	11.5	2.3	18
		% Total	0	0	33.3	12.8	46.2
	3	Count	0	0	1	0	1
		Exp. count	0.1	0.2	0.6	0.2	1
		% Total	0	0	2.6	0	2.6
	Total	Count	2	7	25	5	39
		Exp. count	2	7	25	5	30
		% Total	5.1	17.9	64.1	12.8	100

*Count Expected Count

DISCUSSION

The primary aim of this study was to establish if the BAMIC system is a useful predictor of RTP following hamstrings injury in professional football. The main findings of our study demonstrated a weak but statistically significant association between the grade of HSI on the BAMIC and the RTP. These findings suggest that the BAMIC could provide valuable prognostic information on the RTP which would vindicate its use in the wider management of HSI.

British Athletics Muscle Injury Classification and Return to Play

Our investigation established a weak but statistically significant correlation between the severity of the injury on the BAMIC and RTP. This finding indicates that the BAMIC grading system could be a useful prognostic indicator of RTP from hamstrings strain injuries in professional football.

Our findings are comparable to studies investigating similar outcomes. Biglands *et al.* (2020), reported a significant and indeed stronger correlation, than our findings between

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BAMIC and RTP but this was in a smaller mixed cohort of football and rugby athletes. In track and field studies, Pollock *et al.* (2021) also published a statistically significant increase in RTP between BAMIC grades, except for grades 2 vs 3 and 'a' vs 'b' score. Whilst in professional football, McAuley *et al.* (2022) reported a statistically significant increase in RTP between grades 1a-2c, noting an increase of approximately 3 days in the mean lay-off time for each increase in grade on the BAMIC.

Our study adds to the emerging data to support the use of the BAMIC in predicting RTP following HSI. Our findings demonstrated longer lay-off periods with injuries that extend into the intramuscular tendon of the hamstrings muscle. This injury graded, anatomically as a 'c' on the BAMIC has also resulted in longer RTP timeframes in other studies (Pollock *et al.* 2016; Wangensteen *et al.* 2018; Pollock *et al.* 2021; Shamji *et al.*2021).

Despite these similarities, there appears to be a lack of conformity in RTP timeframes between investigations. Lay-off periods in previous studies range from 33-48 days for a '3c' injury (Pollock *et al.* 2021; Shamji *et al.* 2021) whist our findings demonstrated a mean RTP of 24 days for the same HSI grade. Data from our study indicates only 10% of the variance can be explained by the correlation between BAMIC grade and RTP. Likewise, low variance (7.6-11.9%) was also reported by Wangensteen et al (2018) using a larger mixed-cohort of athletic participants in their analysis of anatomical location (BAMIC) and RTP. They also did not find a statistically significant correlation between BAMIC and RTP (Wangensteen *et al* 2018) contrary to our findings. Although, it is likely to be multifactorial, the variance in the data, raises questions regarding what other confounding factors may influence the RTP and its correlation with injury classification. The BAMIC grading system has been shown to have 'substantial to almost perfect' intra and intertester reliability (Patel *et al.* 2015; Wangensteen *et al.* 2017) suggesting that this is a robust classification system. However, the lack of conformity between return to play timeframes across investigations could be due to variations in medical practices, expertise and demands of the sport in question. Moreover,

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an identifiable limiting factor across all of the published studies in this field highlights that there is a low volume of higher-grade hamstrings injuries available for multivariate analysis. Limited sample sizes across the published field indicates that larger datasets would be required to provide a more accurate model for the prediction of RTP using the BAMIC. Despite demonstrating similar trends, there was insufficient data in our study to accurately estimate timeframes of recovery for each of the BAMIC grades and therefore caution should still be taken when considering RTP timeframes with BAMIC grading.

Time to Return to Full Training and Oedema Length/Cross Sectional Area.

Our study revealed a weak correlation between both oedema length and cross-sectional area on RTP. The BAMIC uses these characteristic measurements of muscle injury to generate the numerical grading of the injury (Pollock *et al.* 2014; MacDonald *et al.* 2019). It would not be unreasonable to anticipate that examples of larger intramuscular oedema or increased cross-sectional area of the muscle affected by the lesion could have a positive correlation with increased RTP, however, our data revealed limited evidence of this.

Contrary, to our findings, Pollock *et al.* (2021) reported a statistically significance association between the length of the oedema and the cross-sectional area of muscle damage on RTP. Analysis of oedema length by Crema and colleagues (2018) revealed no statistically significant association between the oedema length and RTP. Despite these inconsistencies, the findings of a weak correlation within our study advocates further exploration of this as a concept in a larger cohort study.

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British Athletics Muscle Injury Classification and Modified Peetron's Classifications

There was insufficient data to achieve a reliable indication of agreement between BAMIC and Peetron's classification for muscle injuries. Previous larger cohort studies have determined RTP timeframes for HSI based upon the modified Peetron's classification (Ekstrand, Lee and Healy, 2016). The potential transferability of this data to the BAMIC system remains unclear. This provides further justification that larger cohort investigations are required to establish accurate estimates for RTP for each BAMIC grade of HSI to assist medical teams in the management of this injury.

Limitations and Future Recommendations

Our sample population represented the findings at just one professional football club. Although, this may arguably increase validity, with practices in the medical team being more consistent than in a multicentred study, the sample size needs to be considered in the interpretation of the data. This may explain the lack of statistical significance in the relationship between cross-sectional area and oedema length on RTP. Small sample sizes increase the likelihood that random data points can demonstrate a large effect of the overall model (Field 2009). Therefore, care should be taken when interpreting these findings and replication with much larger sample sizes is warranted. However, despite a small population, such data sets potentially provide important applied work undertaken in a professional football environments (Carling, 2017) and provide valuable evidence to inform practitioners. We have presented the confidence intervals around the estimated beta values to provide the uncertainty within the data.

Our outcomes support the findings of McAuley *et al.* (2022) who propose that there needs to be a larger multi-club, multi-league approach to ensure enough HSI are observed. This would provide additional statistical power to manage the variance in coaches, managers,

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and medical practices in management of HSI and would provide a greater insight in the accuracy of the modality in guiding practitioners in making accurate assumptions regarding RTP.

Conclusion

There is currently a paucity of published information on the implementation of the BAMIC for HSI in professional football. This retrospective cohort study has contributed to the evidence-base within this area. The main findings of this study revealed that there was a statistically significant correlation between the BAMIC grade and RTP. These findings would validate the use of the BAMIC as a prognostic indicator of RTP in hamstrings strain in clinical practice. The study also found a weak and statistically non-significant correlation between oedema length and RTP and RTP and cross-sectional area of muscle damage.

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ETHICAL APPROVAL

Ethics approval was granted from the Social Science, Humanities and Law committee at XX, United Kingdom.

DISCLOSURE STATEMENT

The authors report there are no competing interests to declare.

PROVENANCE AND PEER REVIEW

Not commissioned; externally peer reviewed

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Supplementary Material 1

Dataset inputted into regression analyses

Participant	BAMIC (Converted)	RTP (days)	Percentage cross section	Length of oedema (cm)
1	3	6	30	4.9
2	5	5	20	9.6
3	5	8	30	9.9
4	5	6	50	13.8
5	5	14	30	8.7
6	4	24	45	9.4
7	5	39	10	12
8	3	10	40	4.3
9	5	5	0	0
10	5	27	20	4.3
11	5	13	40	11.1
12	2	5	10	0
13	5	15	30	6
14	9	12	70	12.8
15	8	9	0	0
16	2	41	40	3.6
17	5	23	50	9
18	6	22	50	15
19	1	13	50	0
20	5	2	10	3
21	8	56	10	9
22	6	29	30	18
23	5	7	40	12
24	2	5	0	0
25	5	23	25	10.5
26	6	15	0	0
27	5	10	30	9.4
28	9	35	40	9
29	5	8	30	6
30	4	6	60	12
31	8	11	50	12
32	3	5	5	3
33	5	11	80	10
34	0	6	0	0
35	5	19	20	5
36	4	23	40	8
37	6	29	40	10
38	5	34	10	9.5
39	2	11	10	2.8

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Highlights

The key findings from our study were:

- A statistically significant correlation between the grade of HSI using the BAMIC and the TRFT.
- A weak correlation between cross-sectional area of muscle injury and TRFT.
- A weak correlation between length of oedema in the muscle and TRFT.
- These findings would support the use of the BAMIC in the diagnosis of HSI.

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Conflict of Interest file

Conflict of Interest

Any conflicts of interest should be stated here (please refer to **Declaration of Interest** section above). If there are no conflicts of interest please state "None declared".

Ethical Approval

Ethics approval was granted from the Social Science, Humanities and Law committee at Teesside University, United Kingdom.

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